

AI2702 Mobile Robotics Homework 1

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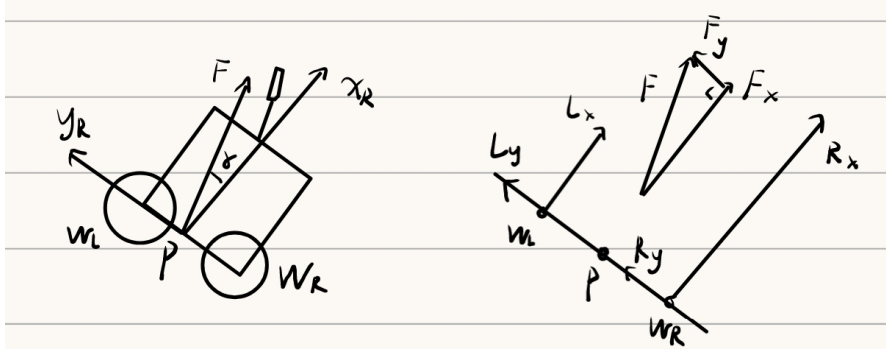


Figure 1: Robot Structure

Let's start from

$$\begin{bmatrix} J_1(\beta_s) \\ C_1(\beta_s) \end{bmatrix} R(\theta)\dot{\xi}_1 = \begin{bmatrix} J_2\dot{\varphi} \\ 0 \end{bmatrix}.$$

However, I find it weird. The constraint seems different than the wheels we know before. At first I tried to model the spherical wheels as parallel steered standard wheels, but I end up with confusion.

There is a severe problem here. Suppose the robot is moving towards direction F with γ from X_R . If there is a difference in speed between the wheels that are parallel, the axis of wheels should break.

Then I consider the case wheels are not parallel. An obvious constraint is that the two wheels always have the same speed on the Y_R direction or the robot explodes. This way, the equations is the same as the case of standard wheels. The coordinate system is defined as in Fig.1. We have $\alpha_L = \frac{\pi}{2}, \beta_L = \beta_1$ for the left wheel and $\alpha_R = -\frac{\pi}{2}, \beta_R = \beta_2$ (yes, index 1 for left and 2 for right):

$$\begin{aligned} r_1\dot{\varphi}_1 \sin \beta_1 &= r_2\dot{\varphi}_2 \sin (\beta_2 - \pi) \\ &= -r_2\dot{\varphi}_2 \sin \beta_2, \end{aligned} \tag{1}$$

This shows that only β_1 is a free control variable.

Then, similar to standard wheels, for spherical wheels:

$$\begin{aligned} [\sin(\alpha + \beta) - \cos(\alpha + \beta)(-l) \cos \beta] R(\theta)\dot{\xi}_1 - r\dot{\varphi} &= 0, \\ [\cos(\alpha + \beta) + \sin(\alpha + \beta)l \sin \beta] R(\theta)\dot{\xi}_1 &= 0, \end{aligned}$$

We use the same simplification as in the textbook to ignore the castor wheel. That is:

$$\begin{bmatrix} \cos \beta_1 & \sin \beta_1 & -l \cos \beta_1 \\ -\cos \beta_2 & -\sin \beta_2 & -l \cos \beta_2 \\ -\sin \beta_1 & \cos \beta_1 & l \sin \beta_1 \\ \sin \beta_2 & -\cos \beta_2 & l \sin \beta_2 \end{bmatrix} R(\theta) \dot{\xi}_1 = \begin{bmatrix} [J_2 \dot{\varphi}]_{2 \times 2} \\ 0 \\ 0 \end{bmatrix}.$$

Simplify the above system of equations by using linear combinations of the first three rows with coefficients $[\sin \beta_1, \sin \beta_2, \cos \beta_1]$ to obtain:

$$[-\sin \beta_2 \cos \beta_2, 1 - \sin^2 \beta_2, -l \sin \beta_2 \cos \beta_2] R(\theta) \dot{\xi}_1 = [r_1 \dot{\varphi}_1 \sin \beta_1 + r_2 \dot{\varphi}_1 \sin \beta_2], \quad (2)$$

and if we substitute equation 1, we get the fourth row. Thus we prove the fourth equation do not add more constraint. In fact, we'd arbitrarily choose the third or the fourth row to calculate inv.

The result is:

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} -\frac{\sin(\beta_1) \tan(\beta_2)}{2} + \frac{\cos(\beta_1)}{2} & -\frac{1}{2 \cos(\beta_2)} & -\frac{\sin(\beta_1 + \beta_2)}{2 \cos(\beta_2)} \\ \sin(\beta_1) & 0 & \cos(\beta_1) \\ -\frac{\sin(\beta_1) \tan(\beta_2) + \cos(\beta_1)}{2l} & -\frac{1}{2l \cos(\beta_2)} & \frac{\sin(\beta_1 - \beta_2)}{2l \cos(\beta_2)} \end{bmatrix} \begin{bmatrix} [J_2 \dot{\varphi}]_{2 \times 2} \\ 0 \end{bmatrix}, \quad (3)$$

Note that since β_2 can be presented with other control variables in equation 1, this expression has many equivalent forms. Also the definition of β_1 and β_2 would cause the result to differ.

If we use the fourth row, the result is:

$$\dot{\xi}_I = R(\theta)^{-1} \begin{bmatrix} \frac{1}{2 \cos(\beta_1)} & -\frac{\cos(\beta_1 + \beta_2)}{2 \cos(\beta_1)} & \frac{\sin(\beta_1 + \beta_2)}{2 \cos(\beta_1)} \\ 0 & -\sin(\beta_2) & -\cos(\beta_2) \\ -\frac{1}{2l \cos(\beta_1)} & -\frac{\cos(\beta_1 - \beta_2)}{2l \cos(\beta_1)} & -\frac{\sin(\beta_1 - \beta_2)}{2l \cos(\beta_1)} \end{bmatrix} \begin{bmatrix} [J_2 \dot{\varphi}]_{2 \times 2} \\ 0 \end{bmatrix}, \quad (4)$$

The matrix inv above is calculated and converted to latex with sympy.